

Faizan Ul Hassan

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Professional Summary

Software Developer with 2+ years building production-grade LLM applications, multi-agent systems, and RAG pipelines. Architected intelligent assistants and semantic search platforms using FastAPI, LangChain, and LangGraph. Proven track record deploying scalable AI solutions with cloud-native infrastructure, vector databases, and CI/CD automation for enterprise clients.

Professional Experience

AI Backend Engineer

Quest

July 2024 – Dec 2025

- **Agentic RAG for Multi-Hop QA:** Engineered agentic RAG system using LangGraph to automate multi-hop reasoning across heterogeneous documents on HotpotQA dataset. Implemented dynamic tool-calling agents with chain-of-thought reasoning and deployed observability tooling via LangSmith CLI.
- **VidVerse:** Architected VidVerse, an AI video intelligence platform supporting multi-format ingestion (direct uploads, YouTube links) with real-time transcription, speaker diarization via Deepgram, and RAG-powered contextual querying using ChromaDB and Supabase backend.
- **Centralized Semantic Search Platform:** Designed centralized semantic search platform aggregating data from multiple heterogeneous sources with custom ETL pipelines, Elasticsearch vector indexing, and FastAPI-based intelligent retrieval endpoints.
- **Multi-Agent Workflow Automation Assistant:** Built multi-agent workflow copilot using LangGraph to automate testing platform operations (user registration, exam scheduling, document management). Integrated RAG assistant and validated performance using AgentEval framework.
- **AI-Powered Courtroom Hearing Portal:** Led backend development of AI courtroom hearing portal with real-time meeting transcription, automated summarization, tagging system, and admin controls for user management, video ingestion, and cron-based newsletter generation with socket-based monitoring.
- **Academic Compliance and Evaluation Assistant:** Developed academic accreditation assistant aligned with UAE standards, utilizing Docling for layout-aware document parsing and Ollama LLMs. Integrated OpenEvals for response accuracy validation.
- **Agentic Workflows with n8n:** Engineered automated workflow pipelines using n8n for RAG-based document processing and task orchestration, integrating Google Drive, Pinecone vector storage, and Gmail for structured business automation.

Python Backend Developer

Codeaza Technologies

July 2023 – June 2024

- **Sentimantle:** Developed backend for an AI-powered sentiment analysis platform for hospitality businesses. Implemented pipelines for 24/7 automated feedback monitoring and served analytics via Flask APIs.
- **Home Schooling Platform:** Engineered an AI-powered education management system. Built RESTful architecture to support personalized progress tracking, student dashboards, and automated reporting insights.
- **Work Book:** Designed server-side logic for an enterprise workflow management system. Streamlined cross-department business processes and team productivity through automated task handling architectures.
- **Keyword Tracking Tool:** Architected an SEO ranking platform with location-based search analysis. Deployed Python automation scripts to scrape precise, ad-free ranking data for strategy optimization.

Education

Bachelor of Science in Computer Science

FAST-NUCES, Islamabad, Pakistan

2019 – 2023

Teaching Assistant: Computer Networks

Technical Skills

- **Languages:** Python, JavaScript (Node.js)
- **Frameworks & Libraries:** FastAPI, Flask, LangChain, LangGraph, Docling
- **LLM & AI Services:** OpenAI, Anthropic Claude, Google Gemini, Ollama, Deepgram
- **Databases:** PostgreSQL, MySQL, MongoDB, Supabase, Elasticsearch, ChromaDB, Pinecone
- **Cloud & DevOps:** AWS, Google Cloud Platform, DigitalOcean, CI/CD, Docker, Git & GitHub, RabbitMQ
- **Tools & Technologies:** n8n, REST APIs, WebSockets, Microservices, RAG Systems, Multi-Agent Architectures